

KSEB Home Electricity Bill Analysis

A Data Analytics Case Study

Kerala State Electricity Board · LT-1 Domestic Tariff · Kondotty Section

January 2025 – March 2026 · 8 Bills · 455 Days · 4,256 kWh · Rs.36,204

Muhamed Jabir

Data Analyst

Microsoft Excel

MySQL

Microsoft Power BI

Executive Summary

This case study presents a data-driven analysis of residential electricity bills from the Kerala State Electricity Board (KSEB) over 14 months from January 2025 to March 2026. The study covers 8 bi-monthly billing cycles totalling 455 days, 4,256 kWh of consumption and a total spend of Rs.36,204. Using Excel for data structuring, MySQL for querying and Power BI for visualisation, this analysis uncovers seasonal patterns, cost drivers, TOD usage behaviour and billing anomalies — with practical recommendations to reduce electricity costs.



Objective

- Identify seasonal consumption patterns
- Understand bill charge components
- Analyse TOD usage: Normal / Off-Peak / Peak
- Detect unusually high bills and find causes
- Provide actionable cost-reduction recommendations

Problem Statement

A 5-member household in Kondotty noticed bills rising significantly. With 3 ACs running in summer, bills were higher but the full extent was unclear. This analysis quantifies the impact, explains seasonal differences and identifies cost-saving opportunities through smarter usage habits.

Data Source

Field	Details
Source	8 Original KSEB PDF Bills
Location	Kondotty Section [6552], Kerala
Tariff	LT-1 Domestic Single Phase
Connected Load	1205 Watts
Meter	Cumulative → TOD (Apr 2025)
Cycle	Bi-Monthly
Period	January 2025 to March 2026
Total	8 Bills 455 Billing Days

Tools Used

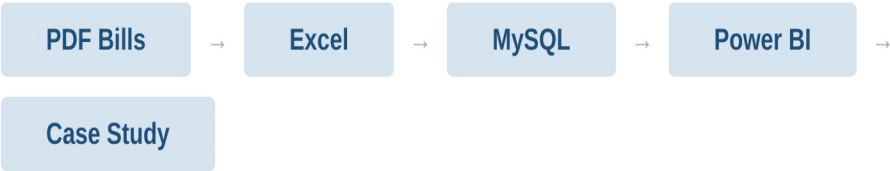
Tool	Purpose
Excel	Data extraction, cleaning, structuring
MySQL	Querying, aggregation, anomaly detection
Power BI	4-page interactive dashboard

Data Preparation & Dataset

Data Preparation Steps

- 1. Extraction:** Manually extracted all values from 8 original KSEB PDF bills.
- 2. Meter Upgrade:** Jan-Feb 2025 = Cumulative meter, no TOD split. Excluded from TOD analysis.
- 3. Short Period:** Mar-Apr 2025 = 31 days due to meter upgrade. Flagged in all analyses.
- 4. Season Labels:** Summer (Mar–May) | Monsoon (Jun–Oct) | Winter (Nov–Feb).
- 5. Excel Structure:** Date, Integer, Decimal (2dp) data types applied correctly.
- 6. SQL Import:** CSV exported as Kseb_home_electricity_bill → MySQL Workbench.

Workflow



Dataset Columns

Column	Type	Column	Type
Billing_Period	Text	Fixed_Charge	Integer
Season	Text	Energy_Charge	Decimal
Bill_Date	Date	FSM/_Refund	Decimal
Billing_Days	Integer	Electricity_Duty	Decimal
Meter_Type	Text	Meter_Rent	Integer
Normal_Units	Integer	Bill_Total	Integer
Off-Peak_Units	Integer		
Peak_Units	Integer		
Total_Units	Integer		

Power BI Dashboard — Page 1: Overview

KPI cards, seasonal bar chart, TOD donut and bill/unit trend line across all 8 billing periods.

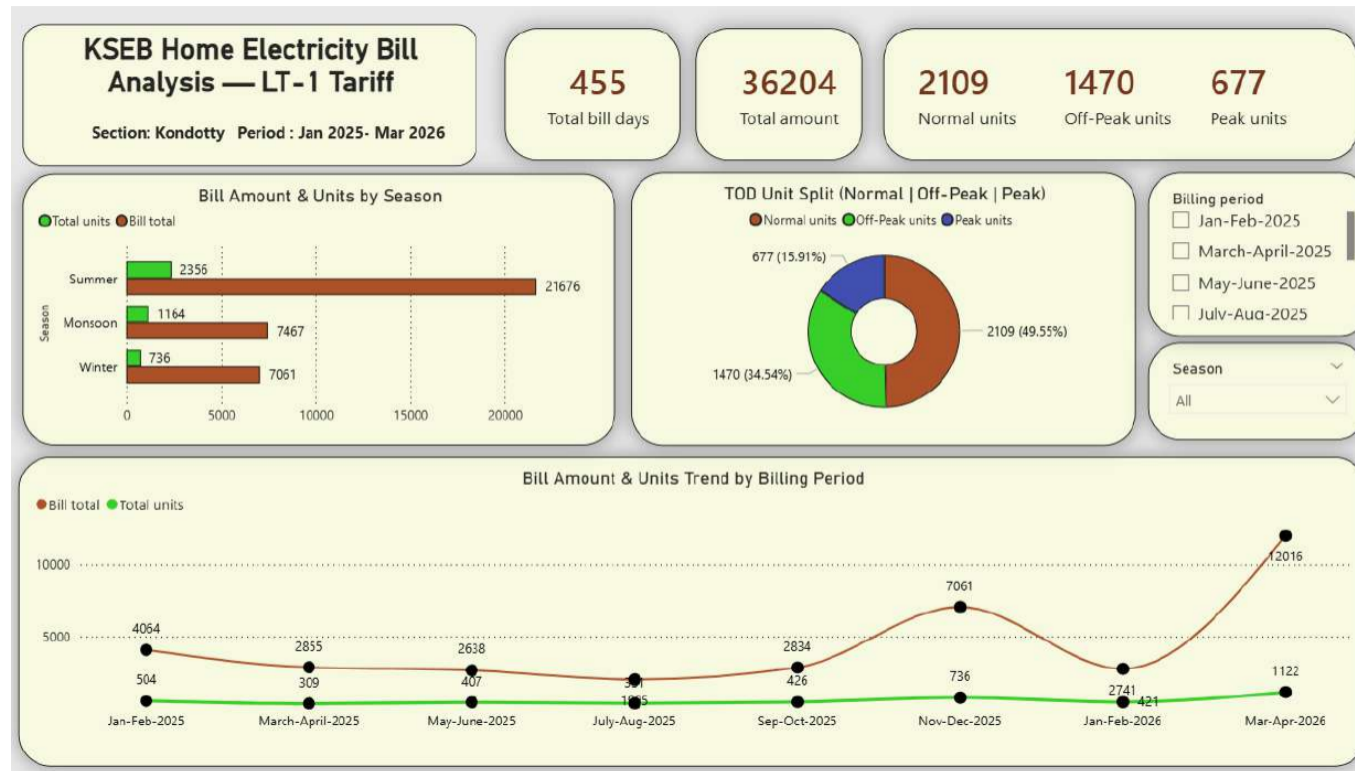


Fig 2: Power BI — Overview Dashboard

- 455 days | 4,256 kWh | Rs.36,204 — all values verified correct
- Monsoon lowest — Kerala climate eliminates AC usage naturally
- Normal 49.55% | Off-Peak 34.54% | Peak 15.91% of total units
- Summer highest: 2,356 kWh and Rs.21,676 — 60% of annual cost
- May 2026 sharp spike to Rs.12,016 — highest bill in 14 months
- Slicers for Billing Period and Season filter all visuals interactively

Power BI Dashboard — Page 2: Consumption & Cost Breakdown

Energy charge KPI cards and clustered bar chart comparing Normal, Off-Peak and Peak units per billing period.

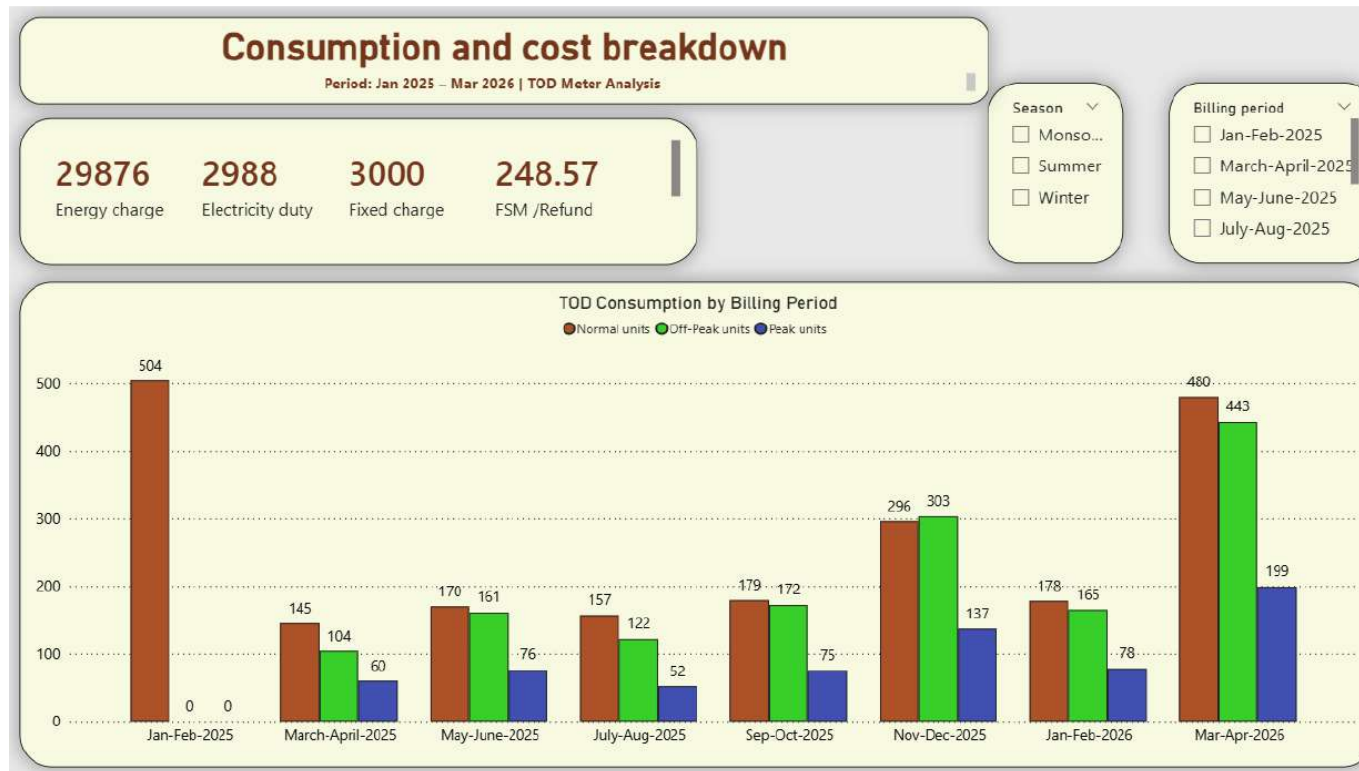


Fig 3: Power BI — Consumption & Cost Breakdown

- Energy charge Rs.29,876 — largest cost at 82.5% of all bills
- Jan-Feb 2025: 504 Normal, 0 Off-Peak, 0 Peak — old Cumulative meter
- Peak hour usage consistently lowest every period — healthy pattern
- Fixed Rs.3,000 + Electricity Duty Rs.2,988 = remaining ~17%
- Mar-Apr 2026 highest: Normal 480, Off-Peak 443, Peak 199 kWh
- Season and period slicers filter all cards and charts together

Power BI Dashboard — Page 3: Detailed Data View

Matrix table with exact values for all columns across 8 billing periods. Conditional formatting on Bill Total.

Billing Period — Detailed Data View									
Billing period	Billing days	Normal units	Off-Peak units	Peak units	Total units	Energy charge	Electricity duty	Fixed charge	Total bill
Jan-Feb-2025	59	504	0	0	504	3301.20	330.12	380	4064
March-April-2025	31	145	104	60	309	2326.40	232.64	240	2855
May-June-2025	60	170	161	76	407	2074.50	207.45	320	2638
July-Aug-2025	61	157	122	52	331	1518.20	151.82	280	1995
Sep-Oct-2025	63	179	172	75	426	2236.00	223.60	320	2834
Nov-Dec-2025	62	296	303	137	736	5888.17	588.82	520	7061
Jan-Feb-2026	59	178	165	78	421	2193.50	219.35	320	2741
Mar-Apr-2026	60	480	443	199	1122	10338.50	1033.85	620	12016
Total	455	2109	1470	677	4256	29876.47	2987.65	3000	36204

Fig 4: Power BI — Detailed Data View

- Bills range Rs.1,995 (Jul-Aug 2025) to Rs.12,016 (Mar-Apr 2026)
 - FSM Refund Rs.-4.21 in Jan-Feb 2026 — KSEB credited overcharged amount
 - Decimals correct: 3301.20, 330.12, 2326.40 etc.
- Energy charge Mar-Apr 2026: Rs.10,338.50 — highest in entire period
 - Total row: 455 days | 4,256 kWh | Rs.29,876 energy | Rs.36,204 total
 - Blue formatting on Bill Total highlights severity of each bill

SQL Analysis — MySQL Workbench

13 queries across 5 sections to validate Power BI findings. Dataset: Kseb_home_electricity_bill.

#	Section	Query	Key Result
Q1	Basic	View all bills chronologically	All 8 bills in correct Bill_Date order
Q2	Basic	Overall totals	8 bills 455 days 4,256 kWh Rs.36,204 ✓ matches Power BI
Q3	Basic	Highest & lowest bill	Highest: Rs.12,016 Mar-Apr 2026 Lowest: Rs.1,995 Jul-Aug 2025
Q4	Consumption	Season wise summary	Summer 2,356kWh Rs.21,676 Monsoon 1,164kWh Rs.7,467 Winter 736kWh Rs.7,061
Q5	TOD	TOD split per period	Normal ~42% Off-Peak ~39% Peak ~18% per period
Q6	TOD	Overall TOD split	Normal 1,605kWh 42.8% Off-Peak 1,470kWh 39.2% Peak 677kWh 18.0%
Q7	Cost	Charge breakdown per period	Energy Rs.29,876 Fixed Rs.3,000 Duty Rs.2,988 FSM Rs.249 Rent Rs.90
Q8	Cost	Charge % of total bill	Energy 82.5% Fixed 8.3% Duty 8.3% FSM 0.7% Rent 0.2%
Q9	Cost	Total of each charge	Grand total Rs.36,204 — matches Power BI exactly ✓
Q10	Comparison	Highest bill per season	Summer Rs.12,016 Winter Rs.7,061 Monsoon Rs.2,834
Q11	Comparison	Lowest bill per season	Monsoon Rs.1,995 Summer Rs.2,741 Winter Rs.7,061
Q12	Comparison	Bills above average (Rs.4,526)	Mar-Apr 2026 Rs.12,016 and Nov-Dec 2025 Rs.7,061
Q13	Comparison	Cumulative vs TOD meter	Cumulative: 1 bill 504kWh Rs.4,064 TOD: 7 bills 3,752kWh Rs.32,140

```

18  -- Overall totals --
19  • SELECT
20      COUNT(*)           AS Total_Bills,
21      SUM(Billing_Days)   AS Total_Days,
22      SUM(Total_Units)    AS Total_Units_kWh,
23      ROUND(SUM(Bill_Total), 2) AS Total_Bill_Amount
24  FROM Kseb_home_electricity_bill;
25

```

Total_Bills	Total_Days	Total_Units_kWh	Total_Bill_Amount
8	455	4256	36204

Fig 5: Q2 — Overall Totals

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35
36  -- SECTION 2 - CONSUMPTION --
37
38  -- Season wise summary --
39
40  • SELECT
41      Season,
42      COUNT(*)           AS Total_Bills,
43      SUM(Total_Units)    AS Total_Units,
44      ROUND(SUM(Bill_Total), 2) AS Total_Bill
45  FROM Kseb_home_electricity_bill
46  GROUP BY Season
47  ORDER BY Total_Units DESC;

```

Season	Total_Bills	Total_Units	Total_Bill
Summer	4	2356	21676
Monsoon	3	1164	7467
Winter	1	736	7061

Fig 6: Q4 — Season Summary

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49
50  -- SECTION 3 - TOD ANALYSIS --
51  -- What percentage of units used in normal, off-peak and peak hours?
52
53  • SELECT
54      Billing_Period,
55      Normal_Units, "Off-Peak_Units", Peak_Units, Total_Units,
56      ROUND(Normal_Units * 100.0 / Total_Units, 1) AS Normal_Pct,
57      ROUND("Off-Peak_Units" * 100.0 / Total_Units, 1) AS OffPeak_Pct,
58      ROUND(Peak_Units * 100.0 / Total_Units, 1) AS Peak_Pct
59  FROM Kseb_home_electricity_bill
60  WHERE Meter_Type = "TOD"
61  ORDER BY Bill_Date;

```

Billing_Period	Normal_Units	Off-Peak_Units	Peak_Units	Total_Units	Normal_Pct	OffPeak_Pct	Peak_Pct
Nov-Dec-2025	296	303	137	736	40.2	41.2	18.6
Sep-Oct-2025	179	172	75	426	42.0	40.4	17.6
Jan-Feb-2026	178	165	78	421	42.3	39.2	18.5
March-April-2025	145	104	60	309	46.9	33.7	19.4
Mar-Apr-2026	480	443	199	1122	42.8	39.5	17.7
May-June-2025	170	161	76	407	41.8	39.6	18.7
July-Aug-2025	157	122	52	331	47.4	36.9	15.7

Fig 7: Q5 — TOD Split

Key Findings

Summer = 60% of Cost	Summer accounts for Rs.21,676 of Rs.36,204 — 60% of annual cost. Driven by 3 ACs for a 5-member household during peak Kerala summer.	TOD Healthy — 16% Peak	Peak hour usage (6PM–10PM) consistently 16–19% across all periods. Not heavily using the most expensive time slot.
Energy = 82.5% of Bill	82.5% of every bill is the energy charge alone. Fixed costs total only Rs.3,090 and cannot be reduced regardless of usage.	Monsoon = Natural Saving	Monsoon bills average Rs.2,489 vs overall Rs.4,526. Kerala climate eliminates AC usage almost completely during June–October.
May 2026: +338% Spike	1,122 kWh in 60 days — 3× normal. Bill jumped to Rs.12,016 from Rs.2,741. All 3 ACs running simultaneously in peak Kerala summer.	TOD Meter — Apr 2025	New TOD meter introduced time-based billing giving full visibility into when electricity is consumed and opening savings opportunities.
Jan 2026: +149% Spike	736 units and Rs.7,061 — a 149% jump from Rs.2,834. Increased winter indoor activity and continuous appliance usage.	Slab Tariff Effect	KSEB charges more per unit as usage rises — bill grows faster than consumption. Why Jan 2026 jumped 149% when units rose only 73%.

Insights & Recommendations

Insights

Consumption

Bills rise in summer due to fans and AC. During monsoon, bills drop naturally as Kerala weather stays cool.

Cost Pattern

82% of every bill is electricity usage charge. Use less electricity, pay less.

TOD Timing

Evening usage (6PM–10PM) is only 16% — good. Running washing machine and water heater after 10PM saves money.

Anomalies

January 2026 was 149% higher due to winter usage. May 2026 recorded 338% surge — 3 ACs for 5 members in peak Kerala summer.

Slab Tariff

KSEB charges more per unit as usage rises — bill grows faster than consumption.

Recommendations & Actions

1. AC Usage — Main Cost Driver

Set all 3 ACs to 24°C (saves ~6%/degree) · Timer after midnight · Stagger rooms · Clean filters monthly · **Rs.200–300 saving per AC per bill**

2. Shift to Off-Peak After 10PM

Washing machine after 10PM · Water heater after 10PM · Charge devices overnight · **Rs.300–400 saving per billing period**

3. Tariff Slab Awareness

Keep bi-monthly usage below 400 units · Do not run all 3 ACs together · Lower slab = lower rate per kWh

4. Monsoon Strategy

Switch off ACs in monsoon · Use fans during cool evenings · **Target: below 300 units in monsoon**

5. Expected Annual Savings

Summer bill: Rs.12,016 → target Rs.7,000–8,000 · **Annual saving: Rs.4,000–6,000**

Conclusion & Project Summary

This case study demonstrates how real household electricity billing data can be transformed into meaningful insights through structured data analysis. By combining Excel for data preparation, MySQL for querying and Power BI for visualisation, a complete picture of electricity consumption and cost patterns was established over 14 months.

Summer months account for 60% of annual spend, energy charges drive 82.5% of every bill and TOD usage is healthy at 16% peak hours. The May 2026 spike of Rs.12,016 — a 338% increase — was caused by 3 ACs running simultaneously during peak Kerala summer for a 5-member household. With targeted actions the household can save Rs.4,000–6,000 annually.

Project Summary	
Project Title	KSEB Home Electricity Bill Analysis — A Data Analytics Case Study
Analyst	Muhamed Jabir — Data Analyst
Contact	jabirmuhamedp@gmail.com linkedin.com/in/muhamed-jabir github.com/muhamedjabir
Period Analysed	January 2025 to March 2026 14 months 8 bills 455 days
Total Units	4,256 kWh across all 8 billing periods
Total Spent	Rs.36,204
Highest Bill	Rs.12,016 — Mar-Apr 2026 (3 ACs in peak summer, 5-member household)
Lowest Bill	Rs.1,995 — Jul-Aug 2025 (Monsoon — natural cooling)
Key Finding 1	Summer accounts for 60% of annual electricity cost
Key Finding 2	Energy charge = 82.5% of every bill — reducing units reduces cost directly
Key Finding 3	TOD peak usage healthy at 16% — shifting to after 10PM saves Rs.300–400/bill
Annual Saving	Rs.4,000–6,000 with AC temperature control and off-peak shifting
Tools Used	Microsoft Excel MySQL Microsoft Power BI
Data Source	8 Original KSEB PDF Bills — Kondotty Section [6552], Kerala